

Our project was on creating a sort of network traffic analyzer that can go a bit beyond sniffing around packets, but also includes other forensic tools that can help us get information on multiple things. Some of the tools included are an ARP scan which helps us find the devices that are on our network by sending ARP requests to each of the IP addresses. We also have a port scan to see which ports are open on our network. It does go through them all one by one, so it takes some time, but it is efficient. One thing we decided to add just for some extra kinda fun was a speed test. What this does is shows you the upload speed, download speed, and ping. Some information you can get from this is that if you have an abnormal speed then it could mean that something is wrong with the network or device that you are on, and you may need to investigate. The next thing we have is an IP-blocking tool. This sniffs for packets and will block IPs that have suspicious traffic which could be classified as something a firewall would detect. The final big thing we have in our network analyzer is logs and packet sniffing. This code helps us capture and analyze the network traffic that is coming into your network. With this, we save all these packets to a .pcap file to make sure we keep records of all the traffic that is coming onto our network. This is what our project includes and some issues we had with creating this code is when we moved it out of our GitHub onto our own device, we initially did not have the correct file path directory for the terminal. So we needed to find the right path, so our code could work. Another thing is, when we were trying to install dependencies, it wouldn't let us, and we also had to figure out how to get the

right path to them. These were the most issues we had with our code and the project overall and the rest of it worked out pretty well.

For our budget, we wanted to still have it be in a reasonable range compared to competitors. So, Solarwinds firewall network analyzer is around a \$4,000 a month subscription or \$48,000 a year. While DataDog is around a \$2,000 a month subscription or \$24,000 a year for around 50 machines. Based on this, we decided to make our budget be based on per machine/host. We decided to have it cost around \$20 per machine and we have some bundle options. One bundle would be 50 machines for \$800 which would be a 20% monthly discount, or you could do 100 machines which would cost \$1600, which is also a 20% discount. We went with this cost because we wanted to make our clientele wider so that we could be appealing to both smaller networks as well as larger networks like companies. We would also say we may have had a budget underrun. We did anticipate our project costing more initially, but we were able to make some budget adjustments and have it be a more affordable option than our competitors.

Our end result for our code, we have many different parts of information that we are given (see screenshots below). First, we have the ARP scan which asks for the target IP or range for the scan, and then it gives back the IP and MAC address. Next, if you choose the port scan, you enter your target IP or hostname, and it gives back the open ports that are on that host. Our third tool is a speed test, which, if you choose this, you get back your download speed and upload

speed in Mbps and your ping in ms. Our biggest tool is the sniff tool that we have, which you can enter a network interface to sniff if you would like, or leave it blank for everything, and then you create a file name for the .pcap file you are saving the packets to. After this, it asks if you want to enable automatic IP blocking, and then it gives you all the packets that it has found. The last thing you can do is exit the program when you are done using the tool. Moreover, I think a lot of the things we learned are being able to use these different security tools in a forensic manner. It really opened our eyes to the multitude of ways you can use a tool that you may not have thought of before. I think we also learned what is most important when creating a forensic tool. You want to make sure it is thorough and gets the job done so the person doing the analysis isn't confused or doesn't need to go somewhere else to get more information. We think that this is really important, especially when trying to create and sell a tool, because people don't want to have to buy so many different subscriptions to investigate one thing. One of the last things we learned is you need to think things through from all different perspectives. When creating a tool like this, you don't know who is going to be using it or who might try to threaten your tool, so you want to make sure you try your best to cover all your bases when it comes to making a versatile tool as well as a secure one.

```
Select Mode:
1: arp
2: port
3: speed
4: sniff
5: exit
Enter mode: 1
Enter target IP or range for ARP scan: 10.10.10.10
IP                               MAC Address
-----
10.10.10.10                      44:b6:be:94:ec:5f
Select Mode:
```

```
Select Mode:
1: arp
2: port
3: speed
4: sniff
5: exit
Enter mode: 2
Enter target IP or hostname for port scan: 10.135.122.177
-----
Scanning target: 10.135.122.177
Scan started at: 2025-04-24 13:17:58.728070
-----
Port 135: Open
Port 139: Open
Press Ctrl+C (or Ctrl+Q) to exit.
```

```
Select Mode:
1: arp
2: port
3: speed
4: sniff
5: exit
Enter mode: 3
Download Speed: 79.05 Mbps
Upload Speed: 94.47 Mbps
Ping Latency: 19.81 ms
```

```
Select Mode:
1: arp
2: port
3: speed
4: sniff
5: exit
Enter mode: 4
Enter network interface for sniffing (leave blank for all):
Enter filename to save captured packets (default: captured_traffic.pcap): testfile
Enable automatic IP blocking for suspicious traffic? (y/n): y
[*] Sniffing started. Press Ctrl+C in the IDLE shell to stop...
[i] UDP packet: 10.135.122.177 -> 138.87.128.1
[i] UDP packet: 10.135.122.177 -> 138.87.128.1
[i] UDP packet: 138.87.128.1 -> 10.135.122.177
[i] UDP packet: 138.87.128.1 -> 10.135.122.177
[i] UDP packet: 10.135.122.177 -> 138.87.128.1
[i] UDP packet: 10.135.122.177 -> 138.87.128.1
[i] UDP packet: 138.87.128.1 -> 10.135.122.177
[i] UDP packet: 138.87.128.1 -> 10.135.122.177
[i] UDP packet: 10.135.122.177 -> 172.217.215.94
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[i] UDP packet: 10.135.122.177 -> 172.217.215.94
[i] UDP packet: 172.217.215.94 -> 10.135.122.177
Select Mode:
```